

Machine Learning: the three examinable topics

Everything needed for Q1–Q3, compressed. Read this only to check yourself after writing from memory — never instead of writing.

TOPIC 1 — WHAT IS MACHINE LEARNING?

Definition. Machine Learning is a process where a computer learns by observing the patterns in data, rather than following hard-coded rules written by a programmer. — coined by **Arthur Samuel, 1959.**

The core flip

TRADITIONAL PROGRAMMING	MACHINE LEARNING
Rules are written by the programmer	Rules are learned from data
Data + Rules → Answers	Data + Answers → Rules (the <i>model</i>)
Fails when rules are too complex to write	Handles problems no human can write rules for
Performance fixed unless code is edited	Performance improves as more data arrives
e.g. GST calculation, °C → °F	e.g. spam detection, face recognition

Characteristics — pick 4

- 1

Learns **from data**, not from hand-written rules.
- 2

Performance **improves with experience** — more data, better results.
- 3

Identifies patterns in data automatically.
- 4

Predicts outcomes for new, unseen data.

Applications — pick 5, one per domain

- **Recommendation systems** — Netflix, Spotify, YouTube
- **Spam detection** — email filtering

- **Medical diagnosis** — detecting disease from X-rays and scans
- **Fraud detection** — banks flagging suspicious transactions
- **Face & image recognition** — phone unlock, security
- **Speech recognition** — Alexa, Siri, Google Assistant
- **Self-driving cars** — reading the road

MARKS LOST HERE

Netflix and Spotify are **one** application (recommendation systems), not two. Spread your five across **different domains**. Never write “etc.” — examiners cannot award marks for it.

TOPIC 2 — THE THREE FORMS OF MACHINE LEARNING

The organising question: what kind of *feedback* does the machine receive while it is learning?

TYPE	FEEDBACK RECEIVED	DATA	GOAL
Supervised	The correct answer is given	Labelled	Predict the output
Unsupervised	No feedback at all	Unlabelled	Discover hidden structure
Reinforcement	Reward or penalty after acting	An environment, no fixed dataset	Maximise total reward

Why three exist: three different real situations. ① The past already contains the answers → supervised. ② There are no answers, you just want to understand the data → unsupervised. ③ There are no answers at all, only outcomes that turn out good or bad → reinforcement.

SUPERVISED

Trained on data where the **answer column is filled in**, so it can fill in the blank for new data.

- **Classification** — output is a category (folder)
- **Regression** — output is a number (scale)
- Spam detection, house prices
- Linear & Logistic Regression, Decision Tree, KNN, SVM

UNSUPERVISED

Only inputs, **no answers**. Finds structure entirely on its own.

- **Clustering** — group similar data points
- **Association** — items occurring together
- Customer segmentation, market basket
- K-Means, Hierarchical, Apriori

REINFORCEMENT

An **agent** learns by trial and error, chasing reward and avoiding penalty.

- Agent · Environment · State · Action · Reward
- Positive RL / Negative RL
- AlphaGo, chess, robotics, self-driving
- Feedback is *delayed* and evaluative

Classification vs Regression — the reliable test

Does a value “in between” two answers make sense?

Yes → **Regression**. No → **Classification**.

QUESTION	IN-BETWEEN VALUE	TYPE
₹10 note vs ₹20 note	a ₹15 note — does not exist	Classification
₹50 lakh vs ₹75 lakh house	₹62.5 lakh — perfectly normal	Regression
Spam vs not-spam	half-spam — meaningless	Classification
20mm vs 30mm rainfall	25.4mm — fine	Regression

THE CLASSIC TRAP

Numeric answers are not automatically regression. Denominations (₹10/₹20/₹50), bus route numbers and star ratings are *categories written as digits* — there is no ₹15 note and no bus 15.5, so they are **classification**.

Also: **Logistic Regression is a classification algorithm** despite its name.

Reinforcement learning components

TERM	MEANING	IN A SELF-DRIVING CAR
Agent	The learner / decision-maker	The car
Environment	The world it acts in	The road and traffic
State	The current situation	Speed, lane, nearby vehicles
Action	What it can do	Steer, brake, accelerate
Reward	Feedback after acting	+1 staying in lane, -1 collision

Positive reinforcement — a reward is added, so the behaviour is repeated (dog sits → treat). **Negative reinforcement** — an unpleasant condition is removed, which also strengthens the behaviour (beeping stops when the seatbelt is fastened).

TOPIC 3 — OVERFITTING & UNDERFITTING

Data is split: **training data** (~80%, the practice questions) and **testing data** (~20%, held back — the real exam). Everything below is about the **gap between the two scores**.

Three students. **A** barely studied — fails practice and fails the exam → **underfitting**. **B** memorised the textbook word-for-word — perfect on practice, fails the rephrased exam → **overfitting**. **C** actually understood it — does well on both → **good fit**.

	TRAINING ACCURACY	TEST ACCURACY	ALSO CALLED
Underfitting	Low ✗	Low ✗	High bias
Good fit	High ✓	High ✓	bias-variance tradeoff
Overfitting	High ✓	Low ✗	High variance

Memory hook. *Over* = fits the training data *too much* → too good on training, bad on test. *Under* = fits it *too little* → bad at everything.

UNDERFITTING

Definition: the model is too simple to capture the pattern in the data. (**High bias**)

Causes:

- Model too simple for the data
- Too few features
- Insufficient training
- Excessive regularisation

Fixes:

- Use a more complex model
- Add more / better features
- Train for longer
- Reduce regularisation

OVERFITTING

Definition: the model memorises the training data including its noise, instead of learning the general pattern. (**High variance**)

Causes:

- Model too complex
- Too little training data
- Trained for too long
- Noisy or irrelevant features

Fixes:

- More training data (*best fix*)
- Feature selection
- Regularisation (L1 / L2)
- Cross-validation, early stopping
- Pruning (trees), Dropout (neural nets)

SUBTLE BUT EXAMINABLE

“More data” fixes overfitting, never underfitting. More data makes memorisation impossible, forcing the model to find the real pattern. But a straight line fitted to curved data stays a straight line no matter how many points you give it — underfitting is a *model capacity* problem, not a data problem.

EXAM TACTICS

- If the question says **“difference between”** — **draw a table**. Faster to write than prose, and examiners reward it.
- Give a **definition first**, then points, then **one concrete example** per concept.
- Never write “etc.” Never write filler such as “in today’s world this is very important.” Neither earns a mark and both cost seconds.
- Naming **Arthur Samuel (1959)** is a free mark. So is naming two algorithms per type.
- 30 minutes, 3 questions → roughly **8 minutes each**, 6 minutes spare for checking.

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Primary source for going deeper: [Google’s Machine Learning Crash Course](#) — the “Framing” and “Generalization” units cover Topics 1 and 3 directly.

Anything here unclear? Ask your teacher — that’s me. Just ask in the chat.